Rank-Metric Codes and Applications to Cryptography

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1 Introduction

Usually, when considering exchange of information between two entities , it is represented as vectors transmitted across a channel.



Figure 1: Representation of an error

In order to detect and correct the error we add some piece of information into our message which is called redundancy.



Figure 2: Representation of an error and correction

In this context, our codewords are vectors, and the errors are vectors with few nonzero coordinates. Those codes are codes in the Hamming metric.They have been widely studies over the last decades.

Now imagine that we have multiple antennas exchanging messages and one antenna completely stops working (Figure 3). We cannot use the decoding in the Hamming metric to correct the error in the message because the error has too many non zero coordinates.

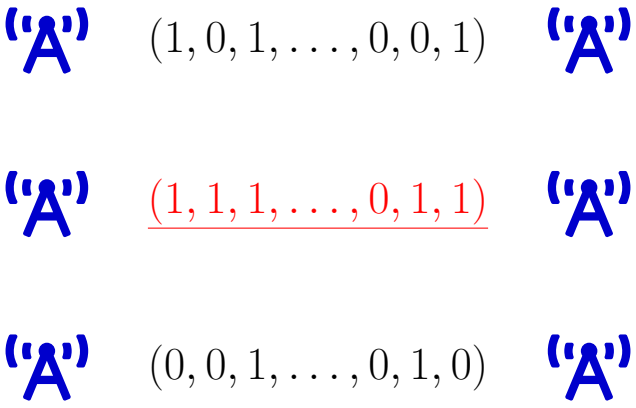
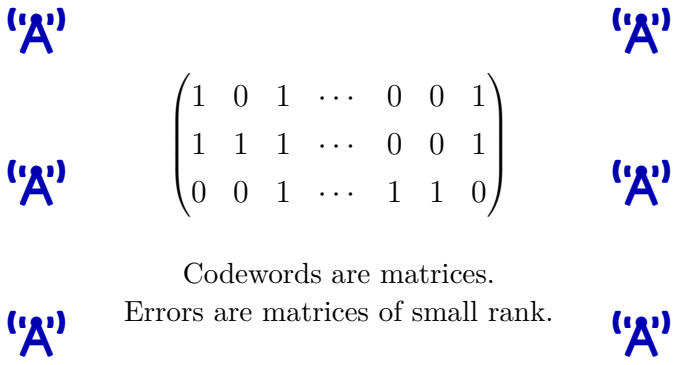


Figure 3: Representation of an error with multiple messages

Alternatively, instead of treating the messages independently, we can group them into a matrix. Consequently, our codewords become matrices and the errors are matrices of small rank. The error in (Figure 3) has a rank of 1, so we will be able to correct it. This is the type of code we will work with. In this report, we will formalize the properties of rank metric codes and explore their applications in modern cryptography



2 Basic Definitions

Classical coding theory can be defined as follows. Let Γ denote a finite-dimensional space over the finite field $\mathbb{K}$. Let d be a distance function over Γ. Then a code is defined as a subset of the metric space (Γ,d). The most common metric space which has been studied is the case $\Gamma = \mathbb{F}_q^n$ with $\mathbb{F}_q$ the finite field of size q, with $\mathbb{K} = \mathbb{F}_q$ and d the Hamming distance.

The goal of coding theory is to find some subset, called codes, which have good properties, such that the minimal distance between codeword is high, or that there exist good decoding algorithm.

Let q be a prime power and $\mathbb{F}_q$ the field of size q. Let m and n be positive integers and denote by $\mathbb{F}_q^{m\times n}$ the $\mathbb{F}_q$ vector space of matrix of size $m \times n$ with entries in $\mathbb{F}_q$.

2.1 Matrix representation

Matrix representation uses the space $\mathbb{F}_q^{m\times n}$of rectangular m $\times$ n matrices over the base field $\mathbb{F}_q$.

For a matrix $A \in \mathbb{F}_q^{m\times n}$, its rank can be defined as the equivalent definitions:

1. The maximum number of linearly independent rows.
2. The maximum number of linearly independent columns.
3. The largest integer r such that at least one minor of order r is non-zero.
4. The dimension of the vector space Im(A).

Definition 2.1. For a matrix $A \in \mathbb{F}_q^{m\times n}$, let rank(A) denote the rank of A. The rank distance between A and B is

$$\begin{aligned} d_r : \mathbb{F}_q^{m\times n} \times \mathbb{F}_q^{m\times n} &\mapsto \mathbb{N} \\ (A, B) &\mapsto rank(A - B) \end{aligned}$$

Proposition 2.1. The rank distance is a distance.

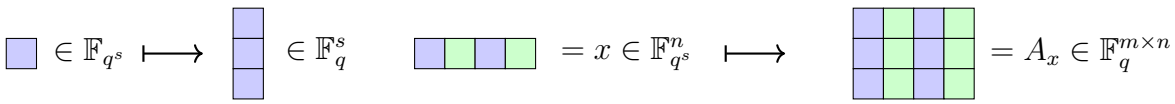
Proof: Let X,Y,Z $\in \mathbb{F}_q^{m\times n}$

1. $d(X, Y) \geqslant 0$ (the rank of a matrix is a positive integer)
2. $d(X, Y) = 0 \iff rank(X - Y) = 0 \iff X - Y = 0 \iff X = Y$
3. $d(X, Y) = rank(X - Y) = rank(Y - X) = d(Y, X)$
4. $d(X, Z) = rank(X - Z) = rank(X - Z + Y - Y) = dim(Im(X - Z + Y - Y))$
   We have $Im(E + F) \subseteq Im(E) + Im(F)$ for E and F two Fq-vector spaces, so $Im(X - Z + Y - Y) \subseteq Im(X - Y) + Im(Z - Y)$.
   And $dim(Im(X-Y)+Im(Z-Y)) \leq dim(Im(X-Y))+dim(Im(Z-Y))$ (by Grassmann's formula).
   So we get $dim(Im(X - Z)) \leq dim(Im(X - Y)) + dim(Im(Z - Y)) \Rightarrow d(X, Z) \leq d(X, Y) + d(Y, Z)$

Definition 2.2. A *matrix code* C in the rank metric is a subset of the space $\mathbb{F}_q^{m\times n}$ endowed with the rank distance. If C is a $\mathbb{F}_q$−-linear subspace of $\mathbb{F}_q^{m\times n}$, then it is a $\mathbb{F}_q$-*linear* code.

2.2 Vector representation

Let v be a vector $\in \mathbb{F}_{q^m}^n$. We can see it as a matrix by looking at the mapping from $\mathbb{F}_{q^m}^n$ to $\mathbb{F}_q^{m\times n}$



$$ext_\beta : \mathbb{F}_{q^m}^n \mapsto \mathbb{F}_q^{m \times n}$$

$$\vec{v} = (v_1, \ldots, v_n) \mapsto \vec{A} = \begin{pmatrix} A_{1,1} & \ldots & A_{1,n} \\ \vdots & \ddots & \vdots \\ A_{m,1} & \ldots & A_{m,n} \end{pmatrix},$$

where $\beta = (\beta_1, \beta_2, \ldots, \beta_m)$ is a basis of $\mathbb{F}_{q^m}$ over $\mathbb{F}_q$ and

$$v_j = \sum_{i=1}^{m} A_{i,j}\beta_i, \quad \forall j \in [1, n].$$

This map is an isometry and is bijective. For a vector v $\in \mathbb{F}_{q^m}^n$, the rank of $ext(v)_\beta$ does not depend on the basis we chose. We can multiply our matrix by an invertible matrix to change our basis, so the rank stays the same. Given a rank code in matrix representation we can construct a code in vector representation with the same size and code distance.

The inverse is also possible. Mapping a $\mathbb{F}_{q^m}$-linear code in vector representation into related code in matrix representation results in a $\mathbb{F}_q$-linear code. However, mapping a $\mathbb{F}_q$-linear code in matrix representation into related code in vector representation does not necessarily result in a $\mathbb{F}_{q^m}$-linear code.

Definition 2.3. Let v be a vector $\in \mathbb{F}_{q^m}^n$. The rank distance between u and v $\in \mathbb{F}_{q^m}^n$ is
$$d_r(u, v) = rank(ext(u - v))$$
It is a distance on $\mathbb{F}_{q^m}^n$. The weight $w_r(v)$ of v is the rank of v.

Definition 2.4. A *vector code* C in the rank metric is a subset of the space $\mathbb{F}_{q^m}^n$ endowed with the rank distance. If C is a $\mathbb{F}_{q^m}-$linear subspace of $\mathbb{F}_q^{m \times n}$, then it is called a $\mathbb{F}_{q^m}$-*linear* code.

2.3 Minimum rank distance

Definition 2.5. Let C be a code in rank metric. The minimum rank distance of C is
$$d = \min(d_r(u, v) \mid u, v \in C, u \neq v)$$

For a linear vector code $\in \mathbb{F}_{q^m}$ of length n, dimension k and distance d, we say that C is a $[n, k, d]$ code. For a linear matrix code $\in \mathbb{F}_q^{m \times n}$ of length $m \times n$, dimension k and distance d, we

say that C is a $[n \times m, k, d]$ code.

Example: Let $q = 2, m = 2$ and $n = 2$. Let $\mathbb{F}_{2^2}$ be the field of order 4, and let $\alpha \in \mathbb{F}_{2^2}$ such that $\alpha^2 + \alpha + 1 = 0$. We define the code C as $C = \langle 1, \alpha \rangle \in \mathbb{F}_2^2$. Then $C = \{(0,0), (1,\alpha), (\alpha, \alpha+1), (\alpha+1, 1)\}$.
We want to calculate the minimal distance of C, let's transform each codeword in a matrix in $\mathbb{F}_{q^m}$ using the basis $(1, \alpha)$.

c	$ext_\beta(c)$	rank
$(0,0)$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$	0
$(1,\alpha)$	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	2
$(\alpha, \alpha+1)$	$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$	2
$(1+\alpha, 1)$	$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$	2

The rank distance of our code is 2.

2.4 Some properties on linear codes

From now on, we will study linear vector codes, which are easier to work with thanks to linear algebra. We can define like in the Hamming metric a generator matrix and a parity check matrix for our codes.

A generator matrix of C is a matrix $\in \mathbb{F}_{q^m}^{k \times n}$ where the vector space is generated by the rows is C. A parity check matrix of C is a generator matrix for the dual of C, where the dual of C is $\{x \in \mathbb{F}_{q^m}^n \mid \langle x, y \rangle = 0 \quad \forall y \in C\}$.

Proposition 2.2. A linear code has a minimum rank distance $d = min_{u \in C, u \neq 0}(rank(u))$

Proof: $d(C) = \min(d_r(u,v) \mid u, v \in C, u \neq v)$ Let u,v $\in C$ such that $u \neq v$ and $d_r(u,v) = d$. Then $rank(u - v) = d$ and C is a linear code hence $u - v \in C$. $\forall c \in C, rank(c) = d_r(c, 0) \geq d$.

Vector codes are more "compact" than matrix code.

Proposition 2.3. Let q=2. Let C be a linear code $\in \mathbb{F}_{q^m}$. Let $C' \in \mathbb{F}_q^{m\times n}$be the matrix code of C by extending each codeword in C with any basis. The number of bits to store a generator matrix of C' is higher than the number of bits to store a generator matrix of C.

Proof: (We suppose that we want to store the entire generator matrix G) Let G be a generator matrix for C, then G has k rows and n columns and its coefficients are $\in \mathbb{F}_{q^m}$. The number of bits to stock G is mnk. C' is a code that live in the F_q-linear space $\mathbb{F}_q^{m\times n}$. Then the length n of C' is $m \times n$. C' has $q^{k'}$ codewords where k' is the dimension of C' over $\mathbb{F}_q$. C is a $\mathbb{F}_{q^m}$-linear space of dimension k so C has $(q^m)^k$ codewords. The number of codeword of C' is the same as C. Therefore we have $q^{mk} = q^{k'}$, $k' = mk$. Let G' be a generator matrix of C'. Then G' has mk rows and mn columns, the number of bits to store G is m^2nk.

We can see that we gain a factor m by working with vector codes. **From now on, we will only work with vector linear codes**

Proposition 2.4. We can correct error of rank $\leq \lfloor \frac{d-1}{2} \rfloor$ i.e $\forall y \in \mathbb{F}_{q^m}^n$ there exist at most one codeword $c \in C$ such that $d_r(y, c) \leq \lfloor \frac{d-1}{2} \rfloor$.

Proof: Let c' be a codeword such that $d_r(y, c') \leq \lfloor \frac{d-1}{2} \rfloor$, the by triangular inequality we obtain $d(c, c') \leqslant d(c, y) + d(y, c') \leqslant d - 1$, so c=c'. This is something which does not depend on the metric.

We want to be able to correct a lot of errors, so we want to maximize the minimum distance d of our codes. We also want to be able to encode a lot of information, so we want the dimension of our code to be high. Unfortunately, like the Hamming metric, for a certain length n of our code, we have to find a balance between k and d.

We have the singleton bound:

Proposition 2.5. Let C be a $[n, k, d]$ code and $n \leq m$, then $d + k \leq n + 1$ .A code achieving this bound is called a MRD code (maximum rank distance).

Proof: Let w_h denote the weight in the Hamming metric i.e the number of non-zero coordinates. Let $c \in C$ and $w_h(c) = a$. Then a has n-a zero coordinates. By extending c into a matrix $\in \mathbb{F}_q$, the matrix has n-a null columns and has rank $\leq a$. So $rank(c) \leq w_h(c)$ and $w_h(c) \leq n - k + 1$ (singleton bound for Hamming metric).

Remark: If $n \geq m$, we extend the code in matrix representation and we transpose each matrix of the code to apply the same process.

3 Linearised Polynomial

Now, we would like to construct [n,k,d] codes where k and d are "big", and with polynomial time algorithm to encode and decode errors. We aim at reproducing in the rank metric existing construction of linear codes in the Hamming metric, known as evaluation codes. In the Hamming metric, the most famous codes about evaluating univariate polynomials are Reed-Solomon codes. However, we cannot use them anymore to construct code, we need to find another type of polynomial.

Linearised polynomial were introduced by Ore in a special case of skew polynomials.

Definition 3.1. Let $a(x) \in \mathbb{F}_{q^m}[x]$ be a polynomial. If $a(x)$ is of the form:

$$a(x) = \sum_{i=0}^{d_a} a_i x^{q^i}$$

then it is called *linearised polynomial*.

The q-degree of a linearised polynomial is defined to be the largest $i \in [0, d_a]$ such that $a_i \neq 0$.

We would like to define the ring of linearised polynomial. However, if we take the usual multiplication on polynomials, then the product of two linearised polynomial may not be a linearised polynomial.

For example, take $P = x^2$ and $Q = x^4$ two linearised polynomials. Then $P \times Q = x^6$ which is not a linearised polynomial. So we have to multiply polynomial in a different way.

Theorem 3.1. *Denote by $\mathbb{L}_{q,m}[x] \subseteq \mathbb{F}_{q^m}[x]$ the set of all linearised polynomials, and by $\circ$ the composition of polynomials. Then $(\mathbb{L}_{q,m}[x], +, \circ)$ is a non-commutative ring with identity.*

Theorem 3.2. *Let $L[x] \in \mathbb{L}_{q,m}[x]$.*

1. *The map $x \mapsto L[x]$ is a linear map over $\mathbb{F}_{q^m}$*
2. *The set of roots of L is an $\mathbb{F}_q$-vector space.*

Proof: (for (1)) Consider $y, z \in \mathbb{F}_{q^m}$ and $\lambda \in \mathbb{F}_q$. Then if

$$L[x] = \sum_{i=0}^{d_L} a_i x^{q^i}$$

$$L(y+z) = \sum_{i=0}^{d_L} a_i (y+z)^{q^i} = \sum_{i=0}^{d_L} a_i (y^{q^i} + z^{q^i}) = \sum_{i=0}^{d_L} (a_i y^{q^i}) + \sum_{i=0}^{d_L} (a_i z^{q^i}) = L(y) + L(z)$$

The second equality holds because the application $x \mapsto x^{q^i}$ is a ring morphism.

$$L(\lambda y) = \sum_{i=0}^{d_L} a_i (\lambda y)^{q^i} = \sum_{i=0}^{d_L} a_i \lambda^{q^i} y^{q^i} = \sum_{i=0}^{d_L} a_i \lambda y^{q^i} = \lambda \sum_{i=0}^{d_L} a_i y^{q^i}$$

The third equality holds because $\lambda \in \mathbb{F}_q$ and $\mathbb{F}_q$ is stable through the Frobenius morphism.

Example 1. Consider $\mathbb{F}_2^3 = \mathbb{F}_2[\alpha]$ with $\alpha^3 = \alpha + 1$ and the polynomial $L[x] = x^4 + x^2 + x \in \mathbb{L}_{2,3}[x]$. The roots of this polynomial are $\{0, \alpha, \alpha^2, \alpha + \alpha^2\}$ and they form a F_2-vector space of dimension 2.

4 Gabidulin Codes

The idea of Gabidulin codes is the same as for Reed-Solomon codes, however we use linearised polynomial instead of normal polynomials.

Definition 4.1. Let $1 \leq n \leq m, x_1, x_2 \ldots, x_n \in \mathbb{F}_{q^m}$ such that $d_r(x_1, x_2 \ldots, x_n) = n$ and note $\alpha = (x_1, x_2 \ldots, x_n)$. Gabidulin Code of length n and dimension k is:

$$G_{n,k}(\alpha) = \{(f(a_1), f(a_2), \ldots, f(a_n)) \mid f \in \mathbb{L}_{q,m}[x], qdeg(f) < k\}$$

Gabidulin codes can also be represented by their generator matrices, which are in the form of a Moore matrix, defined as:

$$M_k(a_1, \ldots, a_n) = \begin{pmatrix} a_1 & a_2 & \ldots & a_n \\ a_1^q & a_2^q & \ldots & a_n^q \\ \vdots & \vdots & & \vdots \\ a_1^{q^{k-1}} & a_2^{q^{k-1}} & \ldots & a_n^{q^{k-1}} \end{pmatrix}.$$

Proposition 4.1. Let $\alpha = (a_1, \ldots, a_n)$ be as in Definition 4.1. Then $M_k(\alpha)$ is a generator matrix of $G_{n,k}(\alpha)$.

Proposition 4.2. Gabidulin codes are MRD and the dual of a Gabidulin code is a MRD code.

4.1 Decoding Gabidulin codes

The structure of Gabidulin codes can help us to make an efficient decoding algorithm. There exist multiple decoding algorithm. The one introduced by Gabidulin is based on the extended

Euclidian division. However, we will focus on another algorithm.

Let $H \in \mathbb{F}_{q^m}^{n-k\times n}$ be a parity check matrix of a [n,k,d] linear code C. Hence, we can correct $\lfloor \frac{d-1}{2} \rfloor$ rank errors. Let $t \in [0, \ldots, \lfloor \frac{d-1}{2} \rfloor]$ and $e \in \mathbb{F}_{q^m}^n$ of weight t. Let $c \in C$ and $y = e + c$. The goal is to find an algorithm in polynomial that gives us c given H and y.

We will do it by finding the error e. The error e is a vector $\in \mathbb{F}_{q^m}^n$ of rank t i.e it has a matrix representation $E \in \mathbb{F}_q^{m\times n}$ of rank t.

Lemma 4.1. *Let $E \in \mathbb{F}_q^{m\times n}$ of rank t. There exist $A \in \mathbb{F}_q^{m\times t}$ of rank t and $B \in \mathbb{F}_q^{t\times n}$ of rank t such that $E = AB$. Moreover, the column space of E is the column space of A and the row space of E is the row space of B.*

Proof: Let A be a matrix whose column form a basis of the column space of E. Then A $\in \mathbb{F}_q^{m\times t}$ and rank(A) = t. Then the column of E must be a linear combinations of the column of A, that is E=AB for some matrix B $\in \mathbb{F}_q^{t\times n}$. Hence, the rows of E are linear combinations of the rows of B. B has a rank at most t and has t rows, B has rank t.

Therefore we can write E = $ext_\beta(e) = A \times B$ with $A \in \mathbb{F}_q^{m\times t}, B \in \mathbb{F}_q^{t\times n}$ If we define a as $ext_\beta^{-1}(A) \in \mathbb{F}_{q^m}^t$

$$e = ext_\beta^{-1}(E) = ext_\beta^{-1}(A \times B) = ext_\beta^{-1}(A) \times B = a \times B.$$

We can verify the second equality by expressing each column of E as a $\mathbb{F}_q$ linear combinations of columns of A and then applicate the $Fq-$isomorphism ext_β^{-1}.

$$\begin{array}{ccc} e & \xrightarrow{Id} & a * B \\ {\scriptstyle ext_\beta}\downarrow & & \uparrow{\scriptstyle ext_\beta^{-1}} \\ A_x & \xrightarrow{Id} & A \times B \end{array}$$

To find e, we will first find such a vector a and then the matrix B.

Both steps can be done with the *syndrome* which is s = $H \times y = H \times (e + c) = H \times e$.

We have to define two polynomial, first the (linearised) syndrome polynomial.

$$s(x) := \sum_{i=0}^{n-k-1} s_i x^{q^i}$$

The other polynomial is $\Lambda \in L_{q,m}[x]$ the span polynomial, which is the polynomial of qdegree t whose roots form the column space of e.

Theorem 4.2.

$$\Omega(x) = \Lambda(x) \circ s(x) \qquad (\text{mod } x^{q^{n-k}})$$

for some $\Omega(x) \in L_{q,m}[x]$ *of q-degree* $\leqslant t$.

Example 2. Consider $F_{2^3} = F_2[\alpha]$ with $\alpha^3 + \alpha + 1 = 0$. α is a generator of the multiplicative group of F_{2^3} which is of order 7. Consider the generator matrix

$$G = (1, \alpha, \alpha^2 + \alpha)$$

As we have seen previously, the code C generated by G has minimum distance 3 and hence can correct 1 rank-error. One parity check matrix of C is

$$\mathrm{H} = \begin{pmatrix} 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha^2 + \alpha \end{pmatrix}$$

An Fq-basis of the column space of e is given by $a = \{\alpha\}$. We consider the error vector $e = (0, \alpha, \alpha) \in (F_{2^3})^3$, which has a rank weight of one.
The error span polynomial is $\Lambda(x) = x \times (x + \alpha) = x^2 + \alpha x$. Its syndrome is $(s_0, s_1) = (H \times e^\intercal) = (\alpha^2 + \alpha + 1, \alpha^2)$.

$$\begin{aligned} \Omega(x) = \Lambda(x) \circ s(x) &= (x^2 + \alpha x) \circ ((\alpha^2 + \alpha + 1)x + \alpha^2 x^2) \\ &= ((\alpha + 1)x^2 + \alpha^4 x^4 + (\alpha^3 + \alpha^2 + \alpha)x + (\alpha + 1)x^2) \\ &= (\alpha^3 + \alpha^2 + \alpha)x \quad (\text{mod } x^2 = x^{q^{n-k}}) \end{aligned}$$

We have $e = a \times B$ with $B = (0, 0, 1) \in {F_2}^3$.

Step 1 : Finding the coefficients of $\Lambda(x)$

We know that Ω is of q-degree less than t, so by calculating $\Lambda(x) \circ s(x)$, we find $\sum_{j=0}^{i} \Lambda_j s_{i-j}^{q^j} = 0$ for $i \in \{t, t+1, \ldots, n-k-1\}$. So we get the linear system of equations:

$$\begin{pmatrix} s_t & s_{t-1}^q & \cdots & s_0^{q^t} \\ s_{t+1} & s_t^q & \cdots & s_1^{q^t} \\ \vdots & \vdots & & \vdots \\ s_{n-k-1} & s_{n-k-2}^q & \cdots & s_{n-k-1-t}^{q^t} \end{pmatrix} \begin{pmatrix} \Lambda_0 \\ \Lambda_1 \\ \vdots \\ \Lambda_t \end{pmatrix} = 0$$

We will find a dimension one solution and any solution will give us the coefficients of the span polynomial. **The complexity of this step is** $O(n^3)$, which is the complexity of solving a linear system with Gaussian eliminations.

Step 2: Finding the roots of $\Lambda(\mathbf{x})$

The error span polynomial being a linearised polynomial, its roots form a $\mathbb{F}_q$ linear space. We have to find a basis of these roots, which will give us our vector a. We have to determine [7]

$$ker(ext_\beta((\Lambda(\beta_0)\Lambda(\beta_1)\dots\Lambda(\beta_{m-1}))))$$

Which will be equal to ext_β(a) of a possible a. **The complexity of this step is** $0(m^3)$

Step 3: Finding the matrix B

Now we need to find the matrix $B \in \mathbb{F}_q^{t\times n}$ such that $e = aB$. Here we solve another linear system using the fact that B is a matrix in $\mathbb{F}_q$. For this we use one more trick, namely that

$$s_i = \sum_{i=0}^{n-1} e_j h_i^{q^i} = \sum_{i=0}^{n-1}\sum_{\ell=0}^{t-1} a_\ell B_{\ell,j} h_j^{q^i} = \sum_{\ell=0}^{t-1} a_\ell d_\ell^{q^i}$$

with

$$d_\ell := \sum_{i=0}^{n-1} B_{\ell,i} h_i. \tag{3.3}$$

We can find $d_0, \dots, d_{t-1}$ by solving

$$\begin{pmatrix} a_0 & a_1 & \dots & a_{t-1} \\ a_0^{q^{-1}} & a_1^{q^{-1}} & \dots & a_{t-1}^{q^{-1}} \\ \vdots & \vdots & & \vdots \\ a_0^{q^{-(n-k-1)}} & a_1^{q^{-(n-k-1)}} & \dots & a_{t-1}^{q^{-(n-k-1)}} \end{pmatrix} \begin{pmatrix} d_0 \\ d_1 \\ \vdots \\ d_{t-1} \end{pmatrix} = \begin{pmatrix} s_0 \\ s_1^{q^{-1}} \\ \vdots \\ s_{n-k-1}^{q^{-(n-k-1)}} \end{pmatrix}$$

Then we have to find B, but $(h_0, \dots, h_{n-1})$ has rank n so we only need to find a change of basis matrix. **The complexity of this step is** $0(n^3)$**, as we again only solve linear system**

The complexity of this algorithm is estimated to be in $0(n^3)$.

Example 3. Consider the example 1, and we are supposing that we don't know the error e. We have to solve the linear system

$$(s_1, s_0^q) \cdot \begin{pmatrix} \Lambda_0 \\ \Lambda_1 \end{pmatrix} = 0$$

So we have $\Lambda_0 \times s_1 + \Lambda_1 \times s_0^q = 0 \iff \Lambda_0 = s_0^q s_1^{-1} \Lambda_1 = (\alpha + 1)((\alpha^2)^{-1}\Lambda_1) = (\alpha)\Lambda_1$. We choose $\Lambda_1 = 1$ and so $\Lambda_0 = \alpha$. The error span polynomial is $\Lambda(x) = (x^2 + \alpha x)$. The roots are $\alpha, 0$ and a basis is given by $\{\alpha\}$. So we take $a = (\alpha)$. Next we solve :

$$\begin{pmatrix} a_0 \\ a_0^{2-1} \end{pmatrix} \cdot (d_0) = \begin{pmatrix} s_0 \\ s_1^{2-1} \end{pmatrix} \iff \begin{pmatrix} \alpha \\ \alpha^2 + 1 \end{pmatrix} \cdot (d_0) = \begin{pmatrix} \alpha^2 + \alpha + 1 \\ \alpha + 1 \end{pmatrix} \iff d_0 = \alpha^2 + \alpha.$$

And we get $d_0 = \sum_{i=0}^{2} B_{0,i}\beta_i \iff \alpha^2 + \alpha = B_{0,0} + \alpha B_{0,1} + \alpha^2 B_{0,2}$. So $B = (0, 1, 1)$ and the error is $e = aB = \alpha \times (0, 1, 1) = (0, \alpha, \alpha)$.

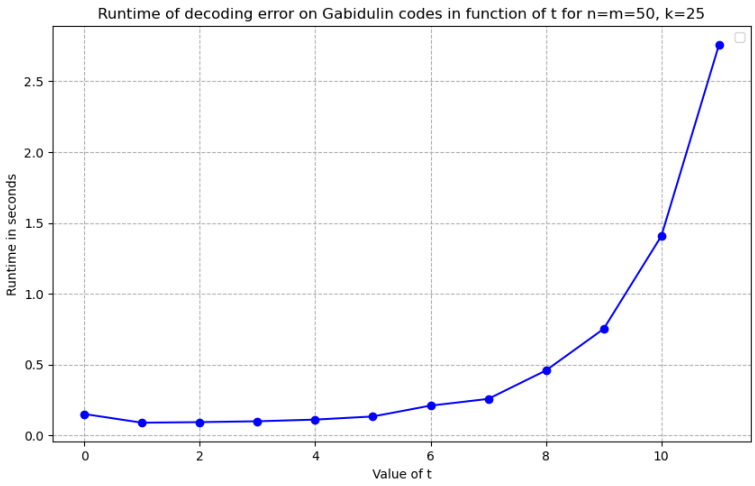
5 Implementations

We have chosen to implement the Overbeck attack and the GPT cryptosystem. However, we have not finished to implement this attack, so we are only presenting the GPT cryptosystem and we will present the attack during the oral part. This attack needs $0(k^2nm^2(s^2 + k))$ operations, plus the operations needed for the Gabidulin code decoding algorithm $(O(m^3 log(m)))$operations .

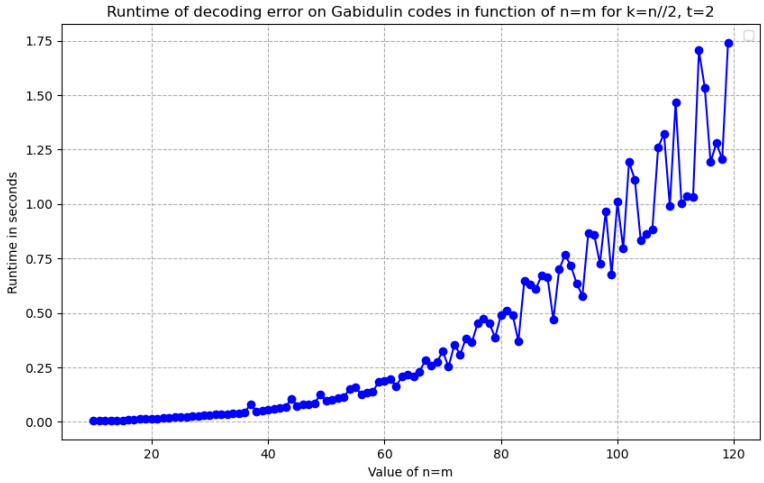
The decoding algorithm is implemented with this document: [7] The attack is implemented using this document [6].

We have tested multiple parameters for n,m,k and t. Most of the time the articles suggest taking n = m, because we need $n \leq m$ and we don't want to work in fields of high order.

We can look at the recommanded parameters to see if we are able to codes with such parameters with a satisfying runtime. We should keep the value of t low, the runtime grow very fast when augmenting the value of t even for small parameters like n=m=50 and k=25.



We can try varying n to see the limit of the length we can use.



For n and m less than 60 and t=2, the runtime of the algorithm is quite low. Here are the recommanded parameters for Hamming metric codes and rank metric codes.

Dec. Complex.	Ham. Met. Gen. Mat.	Rank Met. Gen. Mat.
2^{128}	$[2400, 2006, 58]_2 \approx 100\ KBytes$	$[48, 39, 4]_{2^{48}} \approx 2.2\ KBytes$
2^{256}	$[4150, 3307, 132]_2 \approx 350\ KBytes$	$[70, 50, 5]_{2^{70}} \approx 8.7\ KBytes$

Figure 4: recommanded parameters for a security of 128 bits and 256 bits in Hamming metric and in rank metric
[10]

The parameters for a security of 128 bits are n=m=48 , k=39, t=4. On average, we decode the error on codes with these parameters in 0.001 seconds. For a security of 256 bits, we need n=m=70 , k=50 and t=5. On average, we decode the error on codes with these parameters in 0.0034 seconds.

6 Rank-Metric Code-Based Cryptosystems

One of the main application of error correcting codes outside correcting error is cryptography. Code-based public-key cryptosystems are resilient against all known attacks on quantum computers, unlike RSA and Diffie-Hellman. Hence code-based cryptosystem are good candidates to post quantum cryptography algorithms. Hard problem can be used to design rank-metric codes system. In coding theory, the most known difficult problem is the syndrome decoding problem.

6.1 Hard problems

Syndrome Decoding Problem
Let H be a $((n-k) \times n)$ matrix over $\mathbb{F}_{q^m}$ with $k \leq n$, $s \in \mathbb{F}_{q^m}^{n-k}$ and t an integer. The problem is to find e $\in \mathbb{F}_{q^m}^{n}$ such that $w_h(e) \leq t$ and $He^\intercal = s$ where w_h is the Hamming metric.

This problem has been proven NP-complete, the best known algorithm to solve it are exponential in t. The naive way to solve it is to try all the positions in the error vector and then find the value of the coefficients. However, to find the positions of the error, we have to try $\binom{n}{t}$ combinations.

Rank Syndrome Decoding problem
Let H be a $((n-k) \times n)$ matrix over $\mathbb{F}_{q^m}$ with $k \leq n$, $s \in \mathbb{F}_{q^m}^{n-k}$ and t an integer. The problem is to find e $\in \mathbb{F}_{q^m}^{n}$ such that $rank(e) \leq t$ and $He^\intercal = s$.

This problem is the rank metric version of the Hamming weight syndrome decoding problem, which has been proven to be NP−hard, and possesses the advantage to be resistant to potential quantum computers attack. Unfortunately, the rank metric version of this problem has not been proven to be NP−hard, but the proximity with his Hamming weight version and the decades of

research of this problem, without successful attacks are strongly indicating that this problem may be NP−hard.

6.2 Hamming metric vs Rank metric

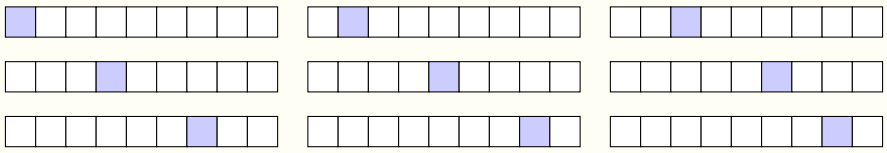
The main drawback of McEliece cryptosystem is the length of the public key. Initially, McEliece proposed to use codes with parameters [1024,524,101], expecting a work factor of at least 2^{65}. The basic attack is to decode by finding k coordinates of the n coordinates hoping that none of the k are in error, and based on this assumption to calculate m. So the parameters had to be high to prevent such an attack.

Recent cryptanalysis lowered the complexity of the decoding algorithm with better attacks. Now, to provide a security of 128 bits, the recommended parameters of the Goppa code used is [2960,2288] and the length of the public key is 1537536 bits. [3]

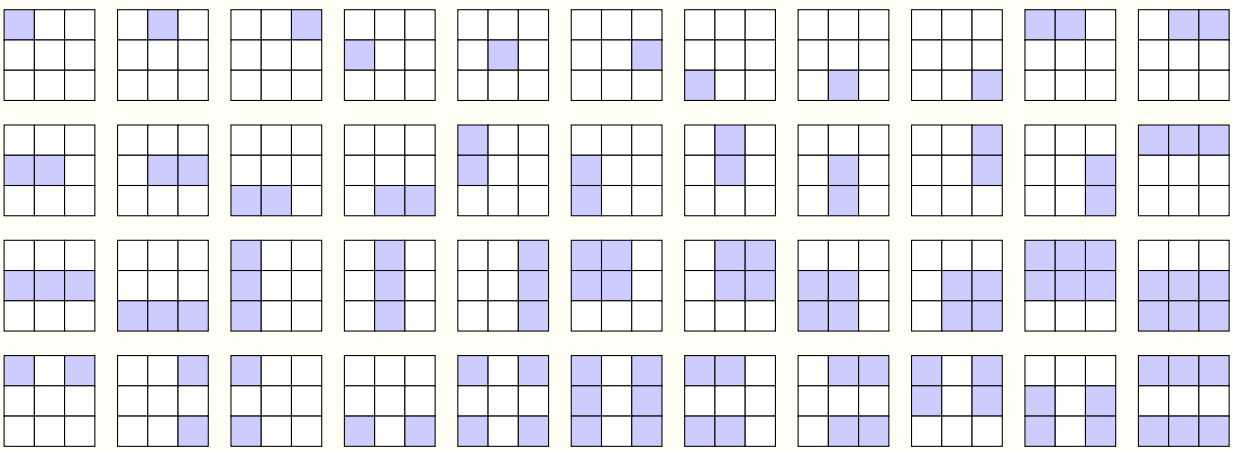
In figure 4 of Chapter 5 the recommendations for 128 bits of security allow us to have public key of 17 000 bits, which is way lower. This is due to the fact that the Rank Syndrome decoding problem is harder to solve that the syndrome problem in Hamming metric. Consequently we can reduce the size of the parameters to obtain a security level.

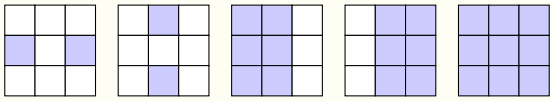
Let's look at an example on how the rank metric decoding problem is harder to solve :

Example 4. Let C be a code in Hamming metric of length 9 and C' be a code in rank metric of length 9. Given a word $c \in C$, if we can correct an error of weight one, here are all the errors we can correct:



Now, with the rank metric, there's much more errors of rank one possible. Here we can correct 49 different errors:





This results in best known attacks having a quadratic term (related to the parameters of the Code) in their exponential coefficients, while for Hamming metric, the best practical attacks have only an exponential term whose exponent is linear in the code parameter. Therefore, decoding a random error by trying all possibilities takes much more time.

GPT cryptosystem

In 1991, Gabidulin published a paper in which he showed a cryptosystem based on the McEliece cryptosystem using Gabidulin codes [1].

Parameters:

q	prime power
m	$m \geq 1$
n	$m \geq n$
k	$k \leq n$

Algorithm 3 GPT Cryptosystem

- **Key Generation:** The receiver chooses a Gabidulin code of minimum rank distance n-k+1 with generator matrix $G \in \mathbb{F}_{q^m}^{k \times n}$. Moreover, they randomly choose $S \in \mathrm{GL}_k(q^m)$ and $X \in \mathbb{F}_{q^m}^{k \times n}$ of column rank at most $s < t$.
- **Private key:** G and S
- **Public key:** $G' = SG + X$
- **Encryption:** Choose a random error vector e of rank weight at most $t - s$ and encrypt the message m as

  $$c = mG' + e = m(SG + X) + e = mSG + mX + e = mSG + e' \mid e' = (mX + e) \text{ rank}(e') \leq t$$

- **Decryption:** Decode c in the Gabidulin code to recover mS. Multiply with S^{-1} to recover m.

An eavesdropper that would get access to c could try to get m in 2 different ways:

- decode the message by finding the error e (solve the syndrome decoding problem)
- exploit the structure of the code

In his article [1], Gabidulin proposed the parameters $n = m = 20$, $k = 12$ and $t = 3$. With those parameters, the size of the public key is n*m*k=4800 which is way better than the length of the McEliece key. He described 2 attacks for which he estimated a work factor of about 2^{100} operations.

In 1995 [2] Gibson published an attack in exponential time on the GPT cryptosystem, making parameters proposed by Gabidulin too weak. However, for higher parameters ($n \geq 30$), the attack took too much time.

In 2008 [3] Overbeck proposed an attack in polynomial time for the GPT cryptosystem, and later for all of its variants. This attack was structural, it was based on the i-th q sum of C which is defined as $\Lambda_i(C) = C + C^q + \ldots + C^{q^i}$ For a random code the dimension of the i-th q sum is larger than the dimension of the i-th q sum of a Gabidulin code with high probability. Hence, Overbeck used this information to perform an attack.

Despite numerous attempts to patch the GPT cryptosystem, the inherent algebraic structure of Gabidulin codes has proven to be an irremediable weakness exploited by attackers.

7 LRPC codes

Definition 7.1 (LRPC code)**.** Let $\mathbb{F}_{q^m}$ be a finite field extension over the finite field $\mathbb{F}_q$. Then we say that a code C [n,k,d] is an LRPC code if its check matrix H (with coefficients hij in $\mathbb{F}_{q^m}$) generates a vector space of dimension at most d over $\mathbb{F}_{q^m}$.We will consider F to be such a vector space, and write F1,..,Fn to be one of its basis. From a vector m $\in \mathbb{F}_{q^m}^k$, we will compute an element c of the code C with c = m.G
With the basics explained, we can introduce the ideas behind syndrome decoding, based on products of vector spaces under small rank conditions.

Definition 7.2 (Matrix A_H^r)**.** Let H := (h_{ij}) a parity check matrix of small rank d over $\mathbb{F}_{q^m}$, with F_1,..,F_n a basis as defined in the above definition and assume that we have v_r $\in \mathbb{F}_{q^m}^n$ a vector of low rank r.Moreover, if there exist a basis E = E_1,...,E_r such that we can write $e_i = \sum_{j=0}^{j=r} Ej * e_{ij}$ and the dimension of vector space FE =$<F_i E_j>_{i\in[1,d]j\in[1,r]}$ is exactly r.d, then we can rewrite the syndrom equations : H.^te=s in equations over $\mathbb{F}_q$ thanks to the fact that we can express e_i in basis E_1,..,E_r, and the syndrom s in basis FE . This means we can find a matrix A_H^r of size n.r x (n-k).d such that $A_H^r.{}^te' =$ s' (where e' and s' are written in basis FE).

Now that we have A_H^r, we do the gaussian reduction of A_H^r and we extract the invertible matrix A_H of rank n*r from the reduction.Once we compute $D_H = (A_H)^{-1}$, we can now directly get the values of e_{ij} and then e itself once we will have determined E. This will form the basis behind the decoding algorithm.

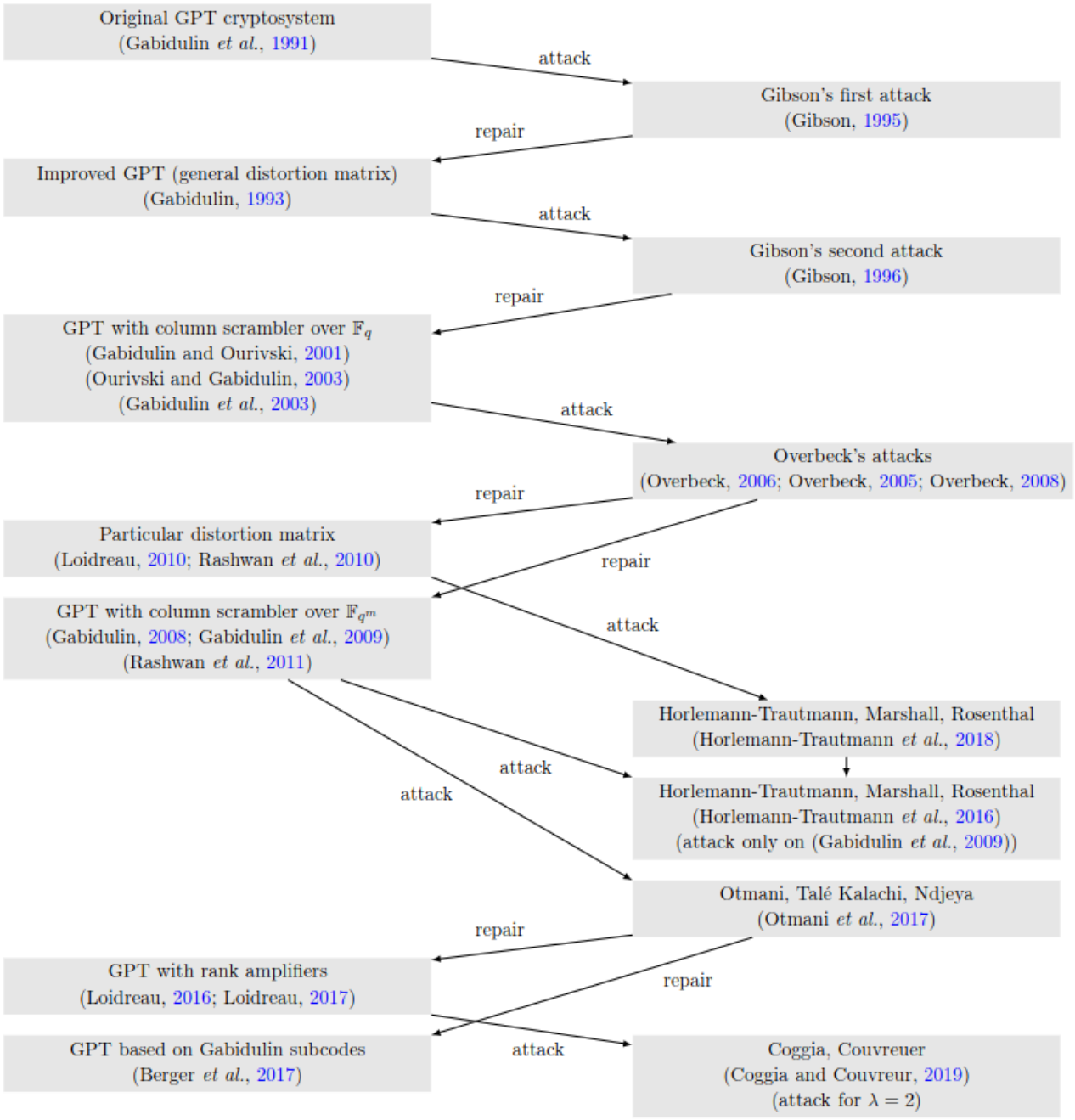


Figure 3.1: Overview of the history of GPT variants and attacks.

[7]

8 Decoding algorithm of LRPC codes

8.1 Decoding Algorithm

- Computing the syndrome & the space it generates:
  to do this we do as always s = H$^{*t}y$ and compute the syndrome space S from $<s_1, \ldots, s_r>$
- Computing E the basis of the error vector
  Since s is the image of e by the multiplication by H, we compute $S_i = F_i.S^{-1}$. Then the intersection of the $S_{i i\in[1,d]}$ is a vector space of basis E=E_1,...,E_r of e over $\mathbb{F}_q$
- Computing A_H^r, getting e We can now construct A_H^r as defined previously, and solve the corresponding system, since we got E in the previous step. This gets us e' that we can rewrite in base F to obtain e.By solving the system m*G = y-e, we get back m.

8.2 More detailed Explanations of the space products method and probabilistic bounds

In **Definition** we assumed we had a small dimension for both the error e and the parity check matrix H, and that it allowed us to compute the cartesian product of the basis obtained from the two vector spaces without further issue. While requiring small dimensions reduces greatly the chances of accidentally reducing the dimension of the cartesian product under r.d, it unfortunately does not make it zero but it is negligible since we will use adapted parameters in cryptographic applications, so we will not consider this case. However, there is a second possibility of failure: if the (n-k) syndroms don't generate the product space EF. Despite this, we have the following bound, that makes the probability of such an event happening exponentially low:

Lemma 8.1 (Bound over the probability of dim EF being strictly less than r.d)**.** *The probability that the n-k syndromes wouldn't generate <EF> is less than* $q^{-(n-k)+r.d}$.

The idea behind the proof is to consider the syndrome space and the n-k>rd associated syndromes: the probability of the second syndrome generating the same space as the first is $\frac{1}{q^{rd-1}}$. The probability of the third syndrome generating the same space as the two previous is at most$\frac{2}{q^{rd-1}}$ (since we can be in the intersection too, so it is at most the sum of the two probabilities).With a proper induction on the number of syndromes, we get that we indeed have at most $q^{-(n-k)+r.d}$ odds of failure.

9 Cryptographic applications

9.1 McEliece setting

The difficulty of solving the RSDP and the resistance to quantum computing threats are making LRPC codes perfect candidates for Mc−Eliece cryptosystem, and unlike Gabidulin codes, the low algebraic structure requirements prevents structure−based attacks like those used on Gabidulin based cryptosystems.

9.2 Description of the setting and use

- Public key: a rank metric code G[n, k] that can decode up to r error, and G'=M.G, where M$\in \mathbb{GL}_n(\mathbb{F}_{q^m})$
- Private key: The above matrix M$\in \mathbb{GL}_n(\mathbb{F}_{q^m})$, and the control matrix H
- Encryption: We compute c = m.G'+e, with e of rank at most r.
- Decryption: we solve the system H.$^t c$ in e and we recover m.

9.3 Comparison with Gabidulin based cryptography and overall pros

The first major advantages compared to Gabidulin cryptosystems is the lack of algebraic structure: all known attacks on LRPC codes are either attacks on generic codes or attacks on LRPC codes which are using circulant matrix (as generator). However using circulant matrix actually is a big advantage for cryptography since those matrix have the form $\begin{pmatrix} Id & A \end{pmatrix}$ and so only giving the matrix A is needed to build the system, thus reducing drastically the size of the key needed. The decoding algorithm is done as quickly in terms of complexity as the Gabidulin one, since matrix inversion in $\mathbb{GL}_n(\mathbb{F}_{q^m})$ can be done in $0(n^3)$ with gaussian reduction. The only real downside would be the fact that the algorithm for decoding is probabilistic, but we saw we could make the probability of failure exponentially small, and since we need to use large keys in the case of cryptographic applications, this is actually not a problem. To put this into perspective, the current NIST recommendation for 128 bits of security using RSA cryptosystem is to use 3072 bits modulus. Using double circulant LRPC codes and with the right parameters we can reach the same level of security with 2809 bits public key and a probability of failure less than 10^{-6}

9.4 Conclusion

Rank metric codes are serious candidates to post quantum cryptosystems due to wide range of error they can correct, which permit low key size. However codes like Gabidulin were the target attacks based on their structure. Despite having been set aside for decades, LRPC-based cryptography has recently gained interest from researchers.

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